A Decision-Support and Algorithmic Roadmapping Framework for the Digital Transformation of Micro-Enterprises: A Review and Conceptual Design

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Abstract

Micro, Small, and Medium Enterprises (MSMEs) in developing and semi-urban economies face substantial structural barriers in adopting digital technologies. While software solutions and digital platforms are increasingly ubiquitous, informal micro-enterprises operate under acute capital constraints, low technical literacy, and profound information asymmetry regarding technological utility. Existing enterprise digital maturity models (DMMs) predominantly presuppose established computing infrastructure and dedicated IT personnel, failing to address the foundational requirements of non-digitized firms. Furthermore, prevailing commercial software offerings promote fragmented, over-engineered tools rather than coordinated, budget-feasible roadmaps. This paper reviews the multidimensional challenges of small-firm digitalization across four thematic pillars: SME maturity models, readiness quantification, combinatorial optimization for IT portfolio selection, and vernacular Human-Computer Interaction (HCI). Building upon this review, we synthesize the theoretical literature to formalize **DigiMitra**, an architectural framework and decision-support system (DSS). DigiMitra models micro-business digital transformation through a four-stage pipeline: empirical readiness diagnosis via supervised machine learning, dynamic feasibility filtering based on human operational capability, capital-constrained intervention selection formulated as a 0/1 Multiple-Choice Knapsack Problem (MCKP), and prescriptive vernacular roadmapping. We conclude with an evaluation methodology based on algorithmic latency, predictive performance, and usability benchmarking.

Keywords: Digital Transformation, Micro-Enterprises, Decision Support Systems, Digital Readiness Index, Multiple-Choice Knapsack Problem, Vernacular AI, ICT4D.

1. Introduction

Micro-enterprises, comprising neighbourhood retail stores (*kiranas*), traditional handicraft workshops, small eateries, and artisanal repair services, form the socioeconomic backbone of emerging markets. In India alone, the MSME sector encompasses over 63 million enterprises. Recent governmental and institutional initiatives have accelerated the adoption of foundational digital rails, most notably low-cost mobile broadband and interoperable digital payments via the Unified Payments Interface (UPI). However, the availability of digital rails has not translated into comprehensive digital transformation.

A critical divergence exists between *technology availability* and *technology fit*. Micro-enterprise proprietors routinely encounter aggressive marketing from software-as-a-service (SaaS) vendors and commercial agencies pushing expensive custom web portals, mobile applications, and enterprise resource planning (ERP) suites. These interventions frequently fail or are abandoned due to three systemic constraints:/

1. **Severe Capital Scarcity:** Micro-enterprises operate on tight operating margins with minimal discretionary technology budgets (typically ₹0 to ₹10,000).

2. **Human Capital and Literacy Barriers:** Proprietors and their staff often possess limited formal technical literacy, making complex software dashboards unintelligible.
3. **Severe Information Asymmetry:** Non-technical merchants lack the domain expertise required to evaluate the expected business utility of distinct digital interventions (e.g., distinguishing whether their business requires a static search listing, a digital catalog, an e-commerce storefront, or automated inventory accounting).

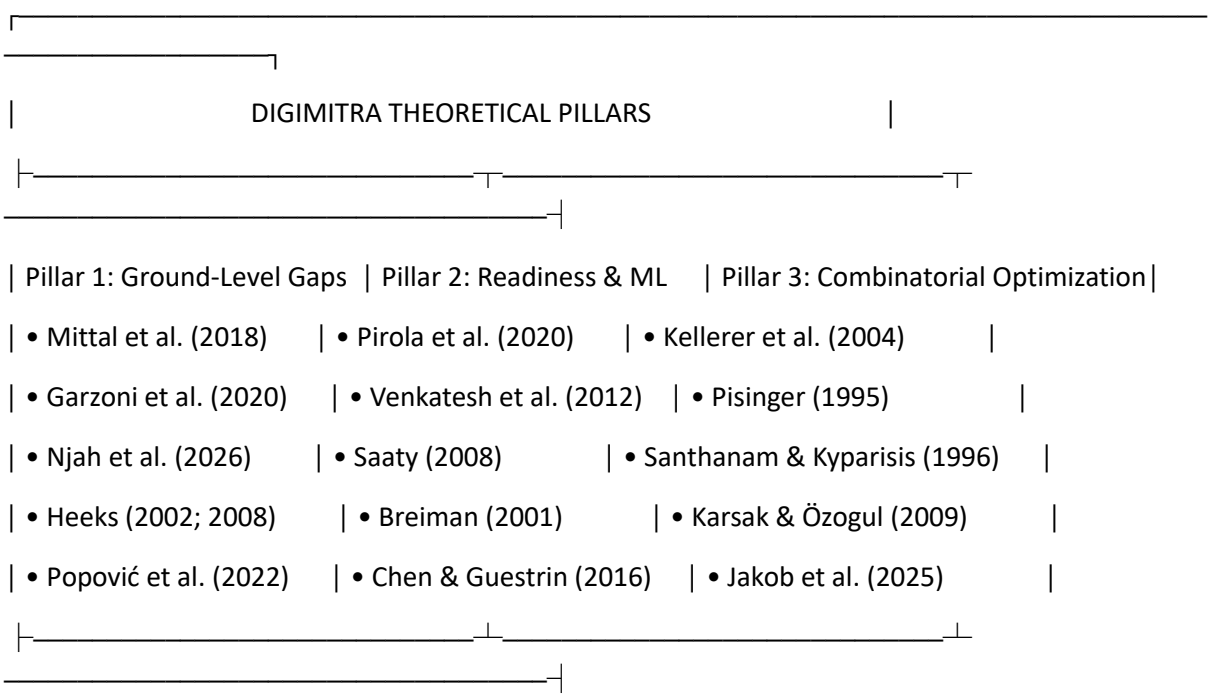
Consequently, technology adoption in this sector is frequently ad hoc, leading to wasteful capital expenditure on uncoordinated tools that fail to generate measurable business value. To address this structural challenge, there is an urgent need for an automated, algorithmic decision-support mechanism that acts as an impartial "digital consultant."

This review paper provides two primary contributions:

- It systematically reviews four foundational pillars of literature: SME maturity frameworks, quantitative readiness modeling, constrained optimization techniques for IT selection, and vernacular interface design.
- It synthesizes these theoretical insights into a formal, end-to-end framework—**DigiMitra**—that models digital transformation roadmapping as a constrained Multiple-Choice Knapsack Problem (MCKP) coupled with supervised machine learning and vernacular explainability.

2. Literature Review: The Four Theoretical Pillars

To construct a rigorous decision-support architecture for micro-enterprises, we examine twenty foundational research studies categorized into four distinct research pillars.



| • Medhi et al. (2007) | Mir & Wani (2025) | Joshi et al. (2020) | Brooke (1996) |
└──┘

2.1 Pillar 1: SME Digital Transformation & Ground-Level Gaps

Conventional digital maturity models (such as CMMI or Industry 4.0 assessment suites) are fundamentally unsuited for informal enterprises. Mittal et al. (2018) conducted a critical review of smart manufacturing maturity models and revealed that virtually all existing frameworks establish a baseline starting state ("Level 1") that assumes pre-existing computerization, digital records, and dedicated IT personnel. For micro-enterprises operating entirely with physical ledgers and cash, this assumption is invalid. Mittal et al. argue for the formal definition of a **"Level 0"** baseline specifically calibrated for non-digitized firms.

Garzoni et al. (2020) analyzed SME digital transformation through the dynamic capabilities perspective, demonstrating that small firms cannot absorb technology through sudden, comprehensive system migrations. Sustainable transformation occurs exclusively through stepped, iterative phases: progressing from basic *Awareness*, to low-risk *Exploration*, and finally to operational *Adoption*. Njah et al. (2026) advanced this concept by developing a multidimensional maturity framework demonstrating that diagnostic evaluations must not terminate at descriptive scorecards; they must map directly to prescriptive, sequential roadmaps.

From a socio-technical perspective, Heeks (2002) established the **Design-Reality Gap** model, proving that information systems deployed in developing economies suffer catastrophic abandonment rates when a severe divergence exists between designer assumptions and the ground-level actuality of user literacy, technical infrastructure, and local institutions. Heeks (2008) further formulated **ICT4D 2.0**, emphasizing that low-income producers require demand-driven, lightweight architectures that solve immediate economic constraints rather than enterprise-grade systems. Finally, Popović et al. (2022) observed that the uncoordinated adoption of disparate digital applications yields zero marginal productivity gains for small businesses, reinforcing the necessity of an orchestrated adoption path to prevent capital fragmentation.

2.2 Pillar 2: Digital Readiness Metrics & Supervised Assessment

To quantify a firm's operational state, researchers rely on structured, multidimensional indexing. Pirola et al. (2020) formulated a digital readiness assessment model for small firms across five core dimensions: Strategy, Technology, People & Skills, Processes, and Integration. Their findings establish that human capability must be weighted alongside physical infrastructure when evaluating readiness.

The human acceptance barrier is formalized by Venkatesh et al. (2012) in the **UTAUT2** framework. UTAUT2 demonstrates that technology adoption in resource-constrained environments is dictated by effort expectancy, facilitating conditions, and habit. In a micro-business context, if an operational tool exceeds the merchant’s cognitive or technical ability, perceived effort rises sharply, causing immediate tool abandonment. To calculate baseline readiness indices, Saaty (2008) introduced the **Analytic Hierarchy Process (AHP)**, which utilizes pairwise comparison matrices to establish objective criteria weightings across competing qualitative indicators.

In predictive modeling over tabular socio-economic data, Breiman (2001) established that **Random Forests**—an ensemble of bagged decision trees—exhibit exceptional robustness against overfitting when evaluating mixed continuous and categorical feature spaces. Chen and Guestrin (2016) developed **XGBoost**, demonstrating that regularized gradient boosting handles sparse tabular features with complex, non-linear feature interactions, providing an optimal baseline for automated readiness scoring.

2.3 Pillar 3: Combinatorial Optimization & Decision Support Systems

Once gaps are identified, selecting an optimal subset of interventions under budgetary constraints represents a classic combinatorial optimization challenge. Santhanam and Kyparisis (1996) proved that information technology portfolio selection must be modeled as a constrained integer programming problem that formally balances budgetary boundaries against projected organizational utility. Karsak and Özogul (2009) successfully operationalized this by coupling multi-attribute decision analysis with mathematical programming to select enterprise software architectures.

In the micro-business domain, interventions within a single technological domain are mutually exclusive (e.g., a merchant cannot adopt a simple WhatsApp Business catalog, a hosted web store, and an integrated enterprise ERP catalog simultaneously in the same phase). Consequently, the decision problem maps precisely to the **Multiple-Choice Knapsack Problem (MCKP)**. Kellerer, Pferschy, and Pisinger (2004) documented the formal proofs, dynamic programming recurrence formulations, and state-space bounds for the MCKP. Pisinger (1995) established minimal algorithms and dynamic programming state-reduction techniques that solve MCKP instances in pseudo-polynomial time. Most recently, Jakob et al. (2025) synthesized modern algorithmic approaches to MCKP variants, demonstrating that dynamic programming approaches achieve sub-millisecond execution times on bounded parameter spaces.

2.4 Pillar 4: Low-Literacy HCI, Vernacular AI & Verification

The interface layer must accommodate users who cannot interact with conventional English-centric software architectures. Medhi, Sagar, and Toyama (2007) conducted pioneering field experiments demonstrating that conventional, text-dense user interfaces fail completely among low-literacy micro-producers; instead, interfaces integrating visual iconography, minimal textual density, and localized speech cues maximize cognitive comprehension. Mir and Wani (2025) examined smartphone affordances among informal merchants in South Asia, concluding that while basic mobile operations (e.g., scanning UPI QR codes) are easily mastered, procedural tasks (e.g., inventory cataloging, configuring domain names) require dedicated vernacular scaffolding.

Joshi et al. (2020) analyzed the state of Indic Natural Language Processing (NLP), demonstrating that over 88% of low-resource language environments suffer from severe structural neglect. Furthermore, they proved that real-world informal trade interactions occur not in high-register formal regional dialects, but through **code-switched language** (e.g., conversational Hinglish). Systems targeting informal merchants must therefore parse and generate conversational code-mixed prompts rather than formal translations. Finally, to rigorously assess the usability of such vernacular interfaces, Brooke (1996) formalized the **System Usability Scale (SUS)**, providing a statistically validated, 10-item instrument for quantifying subjective usability across non-technical user cohorts.

3. Synthesis: The Identified Research Gap

The literature reveals significant progress within isolated domains:

- Information systems researchers have constructed descriptive digital maturity frameworks (Mittal et al., 2018; Pirola et al., 2020).
- Operations research has established exact combinatorial algorithms for constrained resource allocation (Kellerer et al., 2004; Pisinger, 1995).
- Human-Computer Interaction specialists have designed vernacular interfaces for developing markets (Medhi et al., 2007; Mir & Wani, 2025).

The Research Gap: Prior literature does not provide a **unified, closed-loop decision-support pipeline** that simultaneously:

1. Formulates a "Level 0" starting baseline for non-digitized micro-enterprises.
2. Quantifies a standardized Digital Readiness Index (DRI) via supervised machine learning.
3. Imposes a strict human-literacy feasibility gatekeeper on intervention eligibility.
4. Solves a budget-constrained Multiple-Choice Knapsack Problem (MCKP) to maximize transformation utility.
5. Emits a chronological 3-phase roadmap translated into vernacular, code-mixed guidance.

PRIOR ISOLATED APPROACHES:

[Maturity Models] —> Diagnostic Score Only (No budget optimization, no technical roadmap)

[Portfolio Solvers] —> Mathematical Math Only (Assumes enterprise budget and high IT literacy)

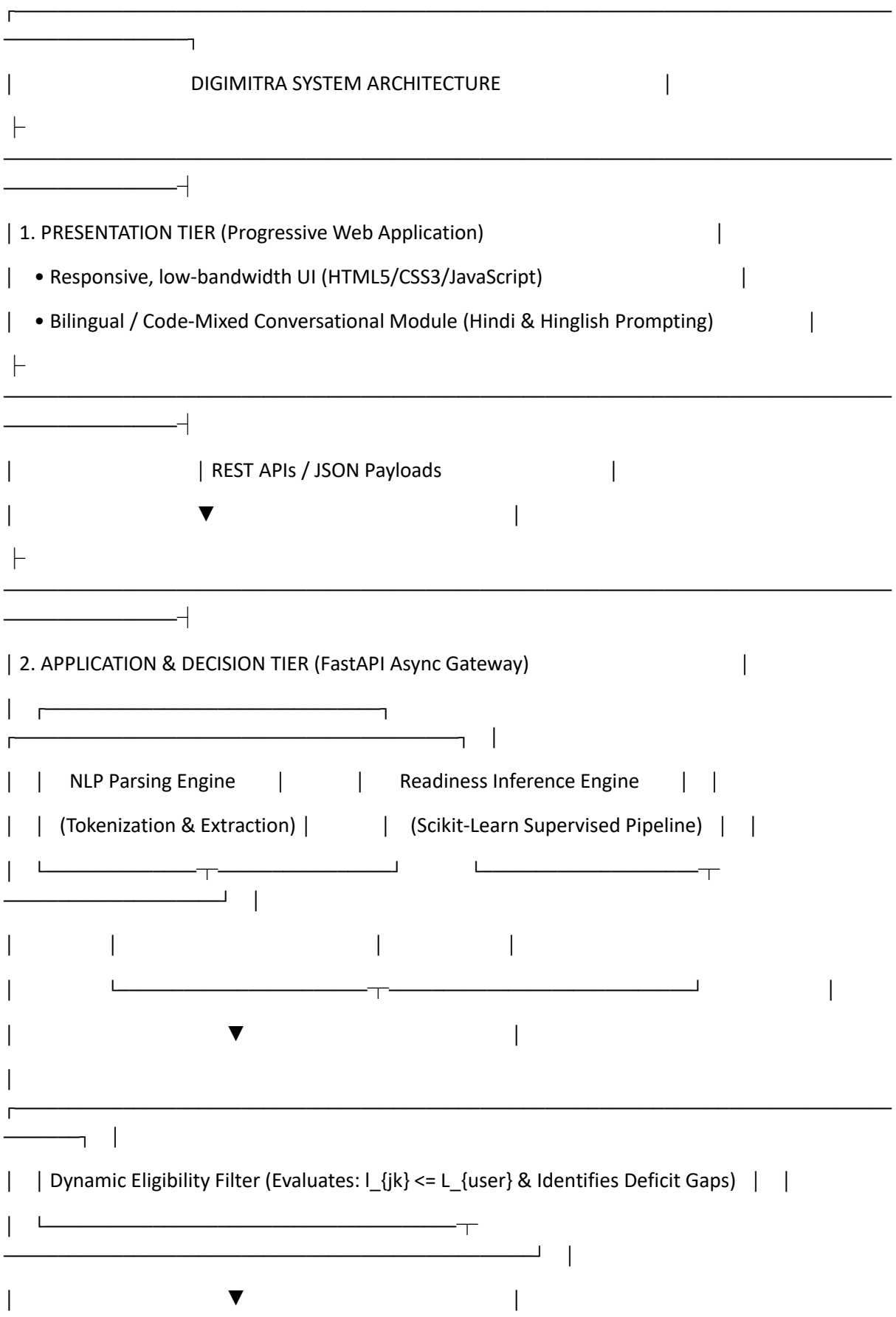
[Vernacular Tools] —> UI Translation Only (Provides single apps without strategic diagnosis)

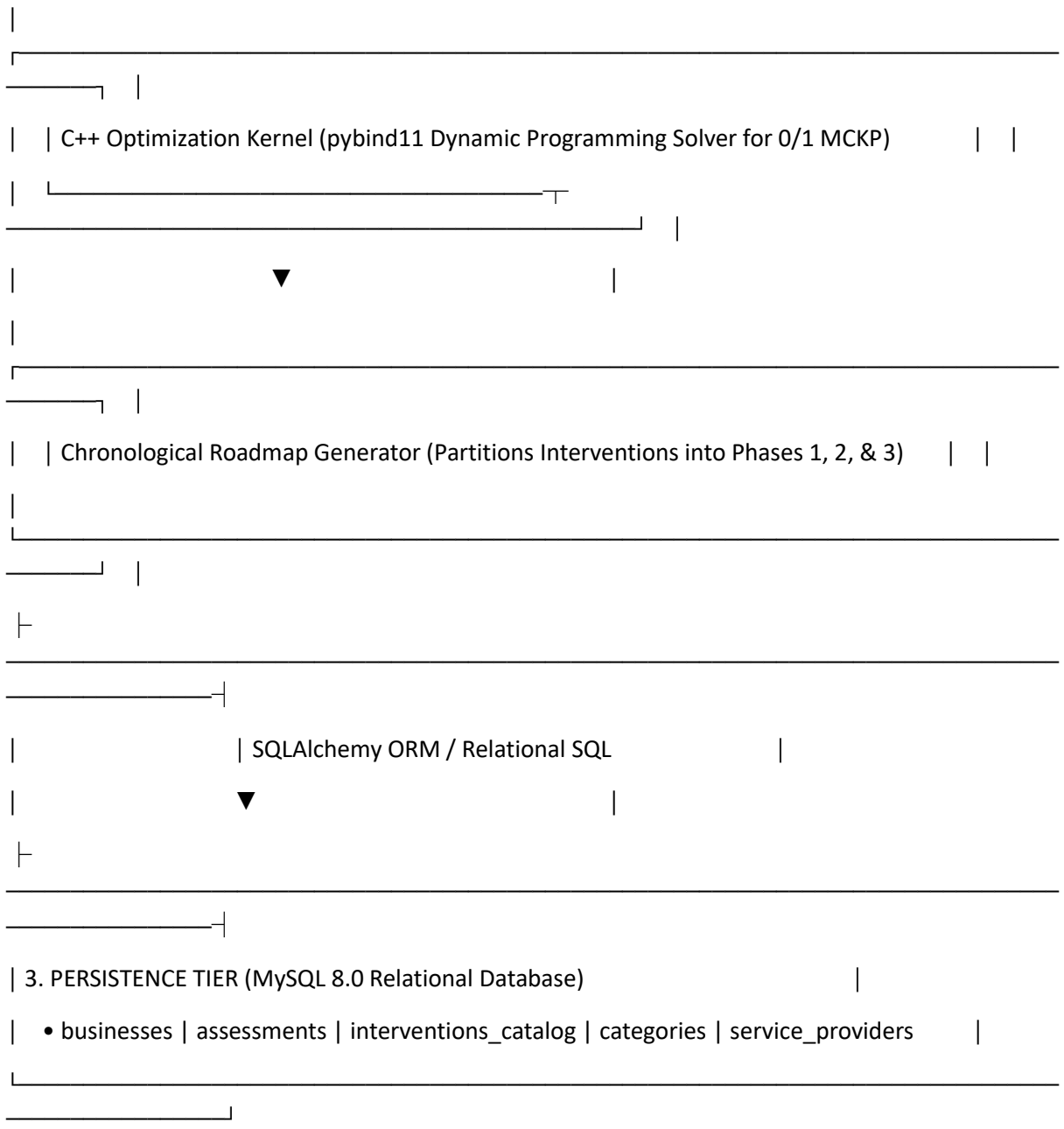
DIGIMITRA UNIFIED PIPELINE:

[Diagnosis (DRI)] —> [Literacy Filter] —> [MCKP Budget Optimization] —> [Phased Vernacular Roadmap]

4. Proposed Conceptual Framework: DigiMitra

DigiMitra is designed as an automated Decision Support System (DSS) to bridge this gap. The platform treats digital transformation not as an event, but as a constrained mathematical optimization problem governed by human capital and financial parameters.


DIGIMITRA SYSTEM ARCHITECTURE
1. PRESENTATION TIER (Progressive Web Application)
• Responsive, low-bandwidth UI (HTML5/CSS3/JavaScript)
• Bilingual / Code-Mixed Conversational Module (Hindi & Hinglish Prompting)
REST APIs / JSON Payloads
2. APPLICATION & DECISION TIER (FastAPI Async Gateway)
NLP Parsing Engine
(Tokenization & Extraction)
Readiness Inference Engine
(Scikit-Learn Supervised Pipeline)
Dynamic Eligibility Filter (Evaluates: l_{jk} <= L_{user} & Identifies Deficit Gaps)



5. Methodology & Mathematical Formulation

5.1 Enterprise Feature Vector & Predictive Readiness Scoring

An enterprise i is represented by a multi-attribute feature vector $\mathbf{x}_i \in \mathbb{R}^d$:

$$\mathbf{x}_i = [\mathbf{v}_i, t_i, m_i, l_i, c_i, p_i, d_i]$$

Where:

- $\mathbf{v}_i$: Business vertical (One-hot encoded: Handicrafts, Kirana, Eatery, Repair Services, Micro-Manufacturing).

- $t_i \in \mathbb{R}^+$: Operational tenure in years.
- $m_i \in \mathbb{R}^+$: Average gross monthly turnover in INR (₹).
- $l_i \in [1.0,5.0]$: Composite digital literacy index calculated from a 4-item Likert assessment (photo creation, messaging, troubleshooting, digital updating).
- $c_i \in \mathbb{R}^+$: Self-declared discretionary technology budget B in INR (₹).
- $p_i \in \{0,1,2,3\}$: Existing payment infrastructure maturity ($0 =$ Cash Only, $1 =$ Static UPI QR, $2 =$ POS Terminal, $3 =$ Payment Gateway).
- $d_i \in \{0,1,2,3,4\}$: Existing digital discoverability score (sum of presence across WhatsApp, Google Business Profile, Social Channels, and Independent Web Domains).

The ground-truth supervisory label—the **Digital Readiness Index (**$\text{DRI}_i \in [0,100]$**)**—is established using an AHP-weighted multi-attribute utility function (Saaty, 2008; Pirola et al., 2020):

$$\text{DRI}_i = \left(w_p \cdot \frac{p_i}{3} + w_d \cdot \frac{d_i}{4} + w_l \cdot \frac{l_i}{5} \right) \times 100$$

where $\sum w = 1.0$ derived from domain consensus ($w_p = 0.30$, $w_d = 0.45$, $w_l = 0.25$). A supervised machine learning estimator (Random Forest Regressor; Breiman, 2001) is trained to map $\mathbf{x}_i \rightarrow \widehat{\text{DRI}}_i$:

$$\widehat{\text{DRI}}_i = f_{\boldsymbol{\theta}}(\mathbf{x}_i)$$

The resulting continuous score assigns the enterprise to one of four maturity tiers:

- **0.0 – 25.0:** Un-digitized / Level 0 Baseline (Mittal et al., 2018)
- **25.1 – 50.0:** Foundational Readiness
- **50.1 – 75.0:** Developing Operational Readiness
- **75.1 – 100.0:** Advanced Digital Maturity

5.2 Combinatorial Optimization via Multiple-Choice Knapsack Formulation

Once diagnostic gaps are identified, the system must select a non-conflicting subset of technological interventions that maximizes cumulative business utility without exceeding the financial capital B or cognitive ability L_{user} of the proprietor.

Let the universal set of technological interventions be partitioned into M disjoint functional categories:

$$J = \{1,2, \ldots, M\}$$

representing distinct capability domains (e.g., $1 =$ Discoverability, $2 =$ Cataloging, $3 =$ Payments, $4 =$ Operations). Each category $j \in J$ contains N_j mutually exclusive implementation tiers:

$$K_j = \{0,1, \ldots, N_j\}$$

where tier $k = 0$ represents the null alternative ($c_{j0} = 0$, $b_{j0} = 0$, $l_{j0} = 0$). Each candidate intervention $k \in K_j$ possesses three empirical parameters:

- $c_{jk} \geq 0$: Capital cost in INR (₹).
- $b_{jk} \geq 0$: Expected business utility score.
- $l_{jk} \in [1.0,5.0]$: Human digital literacy floor required for sustained operational maintenance.

The decision-support optimization is formally defined as:

$$\max_{\mathbf{x}} \sum_{j=1}^{M} \sum_{k=0}^{N_j} b_{jk}\, x_{jk}$$

Subject to:

$$\sum_{j=1}^{M} \sum_{k=0}^{N_j} c_{jk}\, x_{jk} \leq B \quad \text{(Capital Budget Constraint)}$$

$$\sum_{k=0}^{N_j} x_{jk} = 1 \quad \forall j \in \{1, \ldots, M\} \quad \text{(Mutual Exclusivity Constraint)}$$

$$x_{jk} \cdot l_{jk} \leq L_{\text{user}} \quad \forall j \in J,\ \forall k \in K_j \quad \text{(Human Feasibility Gatekeeper)}$$

$$x_{jk} \in \{0,1\} \quad \forall j \in J,\ \forall k \in K_j$$

The human feasibility gatekeeper enforces the UTAUT2 and Design-Reality Gap criteria: any technological intervention whose literacy threshold exceeds the proprietor's measured score ($l_{jk} > L_{\text{user}}$) is pruned from candidate set K_j prior to solving.

The core algorithm is solved using a Dynamic Programming formulation executed in compiled C++20. Let $V(j, w)$ represent the maximum utility obtainable using a subset of categories $\{1, \ldots, j\}$ with total expenditure bound $w \leq B$:

$$V(j, w) = \max_{\substack{k \in K_j \\ c_{jk} \leq w \\ l_{jk} \leq L_{\text{user}}}} \{V(j-1, w - c_{jk}) + b_{jk}\}$$

The algorithmic time complexity is $\mathcal{O}(M \cdot B \cdot \overline{K})$, where $\overline{K}$ is the average number of tiers per category. Given typical micro-enterprise bounds ($M \leq 6$, $B \leq 20{,}000$, $\overline{K} \leq 5$), the C++ implementation

achieves deterministic execution latencies of < 5 ms, ensuring real-time responsiveness within an asynchronous REST gateway.

5.3 Prescriptive Chronological Phasing

The optimal intervention set $\mathbf{X}^* = \{x_{jk} = 1\}$ is partitioned into a 3-phase execution roadmap aligned with the dynamic capability progression of Garzoni et al. (2020):

- **Phase 1: Zero-Cost Foundation ($c_{jk} = 0$):** Immediate configuration of free tools that require minimal operational maintenance (e.g., Google Business Profile verification, WhatsApp Business auto-replies, static UPI QR standees).
- **Phase 2: Local Discovery & Traction (Low-Cost):** Interventions dedicated to regional market visibility and digital cataloging (e.g., localized keyword SEO, digital PDF catalogs, assisted social media listings).
- **Phase 3: Scale & Process Automation (Capital Investment):** Advanced operational deployments (e.g., custom bilingual storefronts, multi-channel payment gateways, Open Network for Digital Commerce [ONDC] seller app onboarding).

6. System Implementation Architecture

The practical realization of the DigiMitra framework is implemented across a modern, modular engineering stack:

Component	Technology	Architectural Role	Literature Justification
Frontend UI	HTML5, CSS3, JavaScript (PWA)	Mobile-first merchant dashboard, visual scorecards, and offline-capable forms.	Medhi et al. (2007): Low-bandwidth, visual iconography for low-literacy users.
Backend API Gateway	FastAPI (Python 3.11+), Uvicorn	Asynchronous REST routing, input validation via Pydantic schemas, ORM interfacing.	High concurrency and rapid serialization of JSON payloads.
Machine Learning	Scikit-Learn, Pandas, NumPy, Joblib	End-to-end data preprocessing pipelines (ColumnTransformer), feature scaling, regression inference.	Breiman (2001); Chen & Guestrin (2016): Reliable inference over mixed tabular features.
Optimization	C++20,	Dynamic Programming solver for	Kellerer et al. (2004);

Component	Technology	Architectural Role	Literature Justification
Kernel	pybind11	the 0/1 MCKP recurrence, exposed as a compiled binary module to Python.	Pisinger (1995): Sub-millisecond execution for combinatorial knapsacks.
Database Tier	MySQL 8.0, SQLAlchemy ORM	Relational schema enforcing referential integrity across business profiles, assessments, and intervention catalogs.	Structured persistence, indexed foreign-key lookups, and district aggregation queries.
Vernacular NLP	Indic-NLP, TextBlob	Parsing code-mixed user feedback and generating conversational explanations in Hindi and Hinglish.	Joshi et al. (2020); Mir & Wani (2025): Code-switched conversational scaffolding.

7. Illustrative Case Study: Sharma Handicrafts

To validate the theoretical framework, consider an illustrative walkthrough representing a typical micro-enterprise in a Tier-2/Tier-3 regional cluster:

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DIGIMITRA DIAGNOSTIC ASSESSMENT REPORT

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ENTERPRISE PROFILE:

Enterprise Name: Sharma Handicrafts Location: Bikaner, Rajasthan

Vertical: Handicrafts & Leather Footwear Tenure: 6.5 Years

Monthly Turnover: ₹45,000 Discretionary Budget: ₹5,000

Proprietor Literacy: 2.3 / 5.0 (Moderate Smartphone Ability; No Technical Desktop Skills)

DIAGNOSTIC READINESS EVALUATION:

Calculated DRI: 38.2 / 100 Maturity Tier: Foundational (Level 1)

Identified Gaps: No online catalog, absence from digital maps, zero web discoverability.

OPTIMIZATION EXECUTION (C++ MCKP DP Solver):

Available Budget Constraint (B): ₹5,000

Max Human Literacy Floor (L_user): 2.3 / 5.0

Pruned Alternatives: Custom E-commerce Store (Req. Literacy: 4.2 > 2.3) -> PRUNED

Cloud ERP Integration (Req. Cost: ₹12,000 > ₹5,000) -> PRUNED

GENERATED 3-PHASE TRANSFORMATION ROADMAP:

[Phase 1: Zero-Cost Foundation | Immediate Execution | Total Cost: ₹0]

• Google Business Profile setup and physical pin verification.

• WhatsApp Business profile creation with basic product catalog (10 items).

• Static UPI QR standee placement at billing counter.

• Vernacular Explanation: "सबसे पहले आपकी दुकान को Google Maps और WhatsApp Business पर रजिस्टर किया जाएगा ताकि स्थानीय ग्राहक आपको आसानी से खोज सकें।"

[Phase 2: Discovery & Traction | Month 1–2 | Total Cost: ₹1,500]

• Digitized smartphone product catalog (PDF/Web-link format).

• Localized SEO keywords targeting regional tourist and handicraft searches.

• Vernacular Explanation: "आपके उत्पादों की एक डिजिटल फोटो सूची बनाई जाएगी जिसे आप ग्राहकों के साथ WhatsApp पर शेयर कर सकते हैं।"

[Phase 3: Assisted Expansion | Month 3–6 | Total Cost: ₹3,500]

• Single-page responsive bilingual showcase landing page.

• Direct WhatsApp ordering integration with UPI payment link generation.

• Vernacular Explanation: "एक साधारण 1-पेज की वेबसाइट शुरू की जाएगी जिससे बाहर के ग्राहक आपके जूते-चप्पल देखकर सीधे WhatsApp पर ऑर्डर दे सकें।"

FINANCIAL & UTILITY SUMMARY:

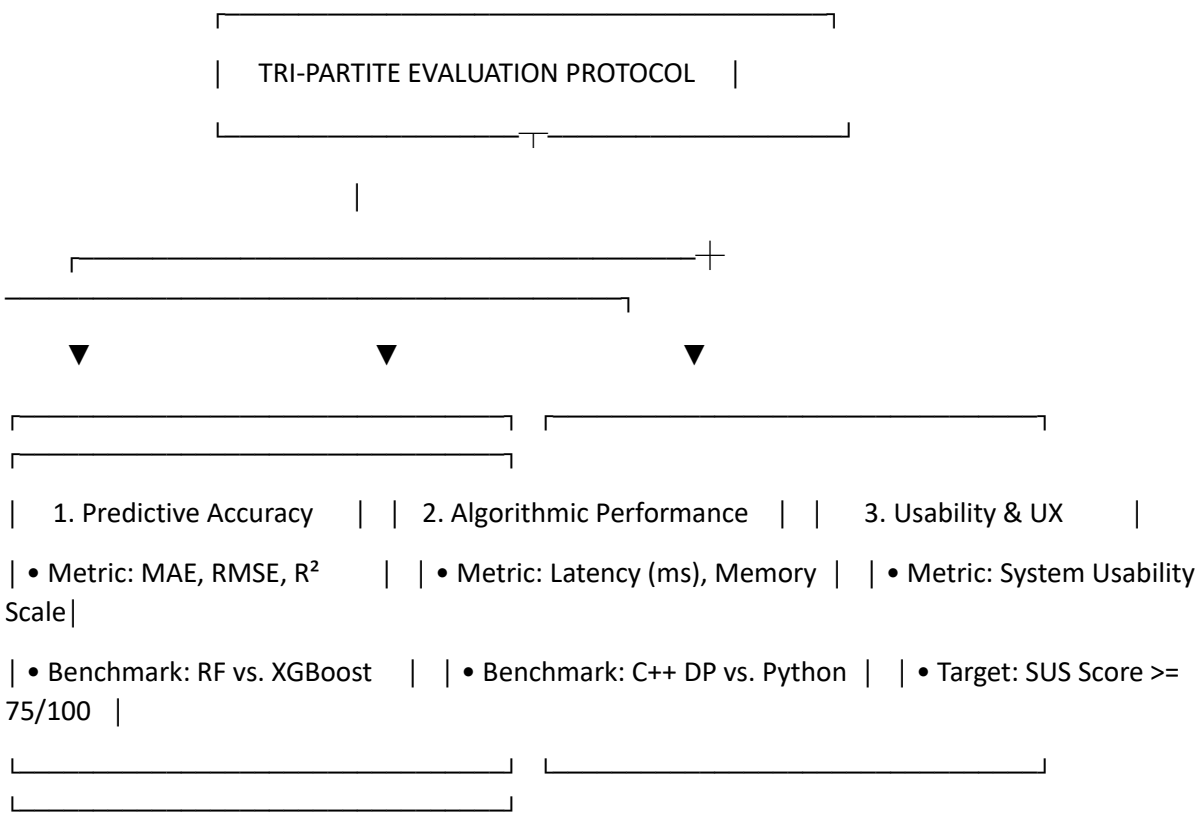
Total Optimized Allocation: ₹5,000 / ₹5,000 (100% Budget Utilization)

Projected Transformation Utility Score: 88.5 / 100

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8. Verification and Evaluation Protocol

To empirically evaluate the DigiMitra framework during deployment, a three-part validation protocol is established:



1. **Predictive Model Evaluation:**
    - Ground-truth synthetic and empirical survey datasets ($N = 300$--500 MSMEs) are evaluated using an 80/20 train-test split and 5-fold cross-validation.
    - Model performance is quantified via Mean Absolute Error (MAE) and Coefficient of Determination (R^2):

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|, \quad R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

    - Target threshold: MAE ≤ 3.0 points on the 100-point DRI scale, and $R^2 \geq 0.95$.

2. **Algorithmic Latency Benchmarking:**
    - Execution latencies of the compiled C++ MCKP Dynamic Programming kernel are benchmarked against an equivalent pure-Python implementation across varying budget scales ($B \in [₹1{,}000, ₹50{,}000]$).

- Target threshold: Execution latency < 10 ms under peak budget constraints, ensuring sub-50 ms total round-trip response time through the FastAPI gateway.

3. **Human Usability & Decision Utility (SUS):**
    - Usability testing conducted with a cohort of non-technical micro-business owners comparing standard IT agency consultation briefs versus DigiMitra's vernacular phased output.
    - Quantified using Brooke's (1996) 10-item System Usability Scale (SUS).
    - Target threshold: A mean composite SUS score $\geq 75/100$, confirming high interface clarity, absence of cognitive overload, and low usability friction.

9. Conclusion & Future Scope

This paper has reviewed the structural challenges surrounding micro-enterprise digital transformation and presented **DigiMitra**, a novel conceptual framework that synthesizes digital readiness assessment, supervised machine learning, combinatorial optimization, and vernacular interaction design. By modeling technological roadmapping as a Multiple-Choice Knapsack Problem constrained by both capital budgets and human digital literacy, the platform eliminates the mismatch between complex enterprise software and the ground realities of informal micro-producers.

Future Scope and Research Directions:

- **ONDC Ecosystem Integration:** Direct API integration with Open Network for Digital Commerce (ONDC) seller applications to automate product cataloging once a merchant crosses the readiness threshold (DRI ≥ 65).
- **Multimodal Voice Conversational Agents:** Expanding the vernacular interface from text-based prompts into speech-to-speech interactive agents (leveraging open Indic speech models) to accommodate fully non-literate merchants.
- **Geospatial District Gap Mapping:** Aggregating anonymized readiness evaluations across municipal jurisdictions to generate real-time district digitalization heatmaps for public policy intervention and micro-lending assessments.

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